Héctor Daniel Díaz-Rodríguez

High Performance Computing | AI | LLMs | Complex Networks | CUDA Programming

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Skills

Computer Scientist and PhD Candidate with research experience in Artificial Intelligence, Complex Networks, High-Performance Computing, and Scientific Software Development. Experienced in designing computational models, optimization methodologies, and large-scale data analysis pipelines. Skilled in Python, C++, CUDA C++, machine learning, and graph analytics, with a strong interest in AI-assisted software development, large language models, and next-generation HPC systems.

Programming Languages and Tools: C/C++, Python, SQL, JavaScript, R, Git, Linux, Jupyter, PyTorch, igraph, NetworkX, NumPy, Pandas.

High Performance Computing: CUDA C++, Parallel Programming, GPU Computing, Concurrent Programming.

Artificial Intelligence: Machine Learning, Deep Learning, Large Language Models (LLMs), Prompt Engineering, Neural Networks, Graph Data Mining, Multi-objective Optimization.

Work Experience

Center for Research and Advanced Studies of the National Polytechnic Institute Sept 2022 - Present

PhD Candidate

- Developed high-performance graph analytics applications using CUDA C++ for large-scale cybersecurity datasets.
- Designed and optimized a GPU-based implementation of the Louvain community detection algorithm using parallel graph traversal and modularity optimization techniques.
- Implemented CUDA kernels for bipartite graph construction, network projection, and backbone extraction, leveraging thousands of GPU threads for accelerated execution.
- Evaluated algorithmic performance through benchmarking experiments, analyzing speedup, scalability, and computational efficiency of GPU implementations.

Selected CUDA Projects **Repository:** diazmx.github.io/pages/cuda

- GPU-accelerated Louvain Community Detection using CUDA C++.
- Parallel Bipartite Graph Construction and Projection.
- GPU-based Backboning Algorithms for Large-Scale Networks.

ITESO University & UVM University Aug 2024 - Present

Adjunct professor

- Taught undergraduate courses in Concurrent Programming, Computer Architecture, Data Structures and Algorithms, Object-Oriented Programming, Relational and NoSQL Databases, and Discrete Mathematics. I received the **Egregius award**, which recognizes excellence in teaching.

Selected Publications

- **ABAC Policy Mining Using Complex Network Analysis Techniques (2025)** Developed a complex network-based approach for ABAC policy mining to automatically extract access control rules from logs; published in a peer-reviewed journal (DOI: `https://doi.org/10.3390/app152312571`).
- **Knowledge modelling for ill-defined domains using learning analytics: Lineworkers case (2020):** Developed a learning analytics–based approach to build knowledge models from textual data in vocational education, using semantic networks and Normalized Web Distance; published in a peer-reviewed conference (DOI: `https://doi.org/10.1007/978-3-030-57799-5_42`).
- **A multi-objective optimization methodology for the joint selection of projection and filtering methods in bipartite networks (2026):** Submitted to Computers & Security (under review). Proposed a multi-objective optimization methodology to select optimal bipartite network projections and backboning techniques for robust network modeling in cybersecurity applications; currently under peer review.

Education

Center for Research and Advanced Studies of the National Polytechnic Institute Sept 2020 - Aug 2026
PhD Candidate & MSc in Computer Science ***CGPA: 9.8/10***

University of Colima Sept 2016 - Aug 2020
B.E. in Computer Science ***CGPA: 9.5/10***